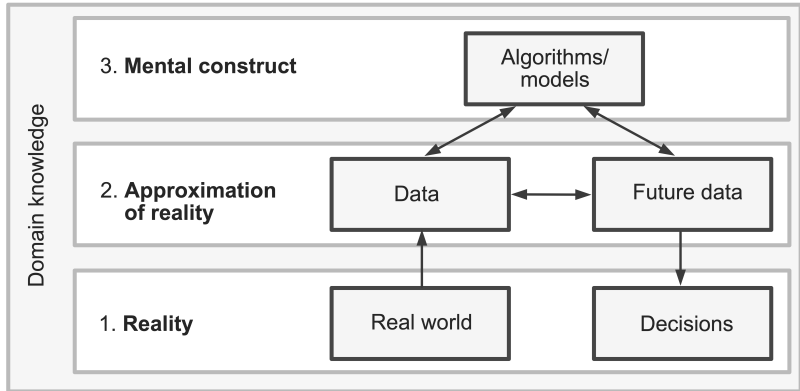


Model diagnostics

Ciaran Evans

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Question: Is our model useful?

Model assessment

We observe data $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$ and fit the model

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}$$

How do we assess whether the model is useful?

Model assumptions

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}$$

What assumptions do we make when fitting this logistic regression model?

Shape assumption

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Shape assumption: The log-odds are a linear function of the explanatory variables (i.e., the systematic component is correctly specified).

Question: How do we assess the shape assumption in a *linear* regression setting?

Residual plot?

Raw residuals: $Y_i - \hat{p}_i$

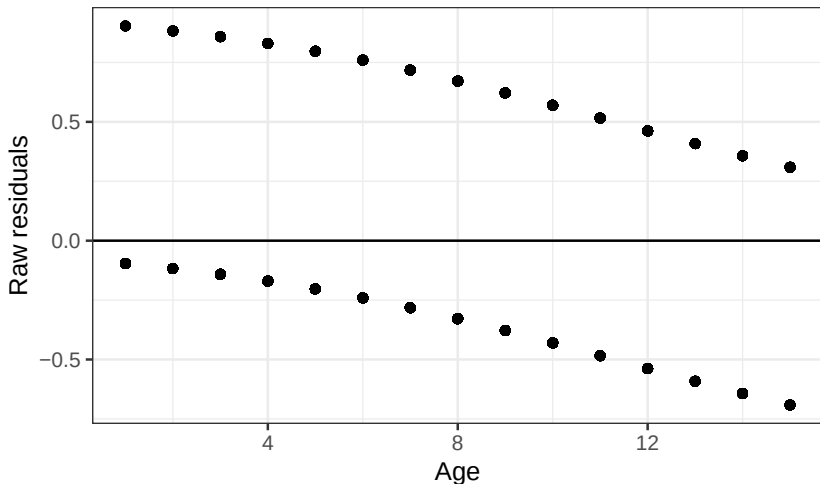
Question: what will the plot look like if we plot $Y - \hat{p}$ vs. X ?

Residual plot?

Raw residuals: $Y_i - \hat{p}_i$

Question: what will the plot look like if we plot $Y - \hat{p}$ vs. X ?

Example: predicting dengue status with Age



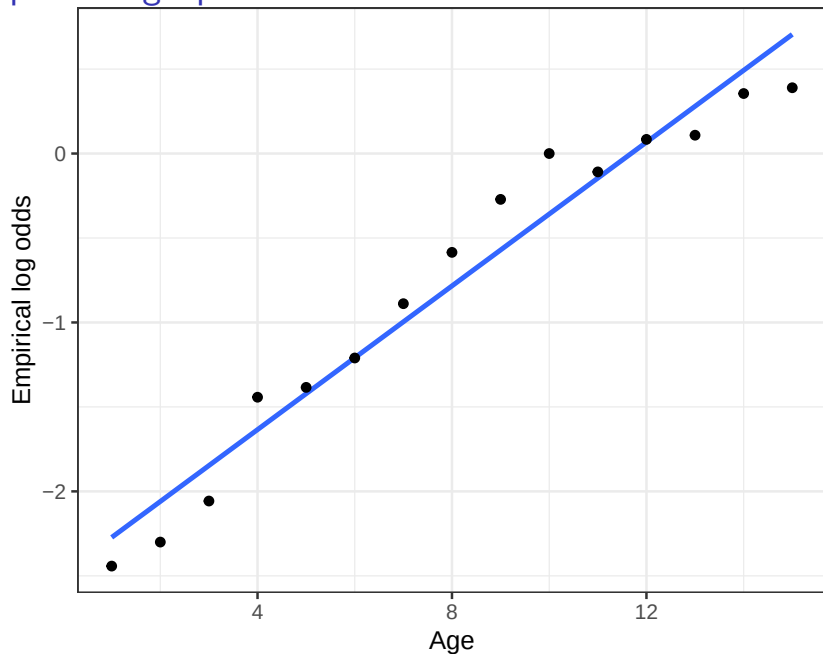
Alternative: empirical logits

For each Age, calculate the *empirical* log odds of dengue (aka *empirical logits*):

```
## # A tibble: 15 x 5
```

##	Age	NoDengue	YesDengue	p	log_odds
##	<int>	<int>	<int>	<dbl>	<dbl>
##	1	1	184	16 0.08	-2.44
##	2	2	419	42 0.0911	-2.30
##	3	3	532	68 0.113	-2.06
##	4	4	457	108 0.191	-1.44
##	5	5	487	122 0.200	-1.38
##	6	6	369	110 0.230	-1.21
##	7	7	360	148 0.291	-0.889
##	8	8	289	161 0.358	-0.585
##	9	9	244	186 0.433	-0.271
##	10	10	185	185 0.5	0
##	11	11	165	148 0.473	-0.109
##	12	12	126	137 0.521	0.0837
##	13	13	96	107 0.527	0.108

Empirical logit plot



Empirical logits with a continuous variable

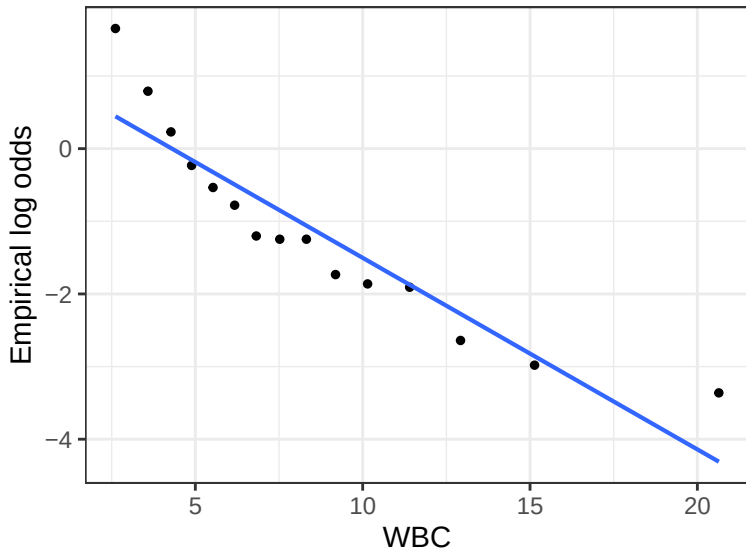
Trying with WBC:

```
## # A tibble: 6 x 5
```

```
##      WBC NoDengue YesDengue      p log_odds
##    <dbl>    <int>    <int> <dbl>    <dbl>
## 1  0.9         0         1     1      Inf
## 2  1.15        0         2     1      Inf
## 3  1.23        0         1     1      Inf
## 4  1.25        0         1     1      Inf
## 5  1.54        0         3     1      Inf
## 6  1.58        0         1     1      Inf
```

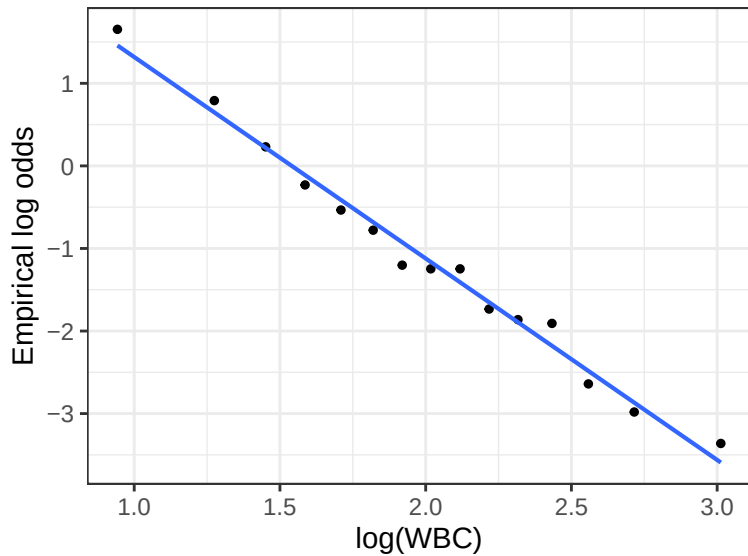
What has gone wrong? What should we do differently?

Binned empirical logit plots

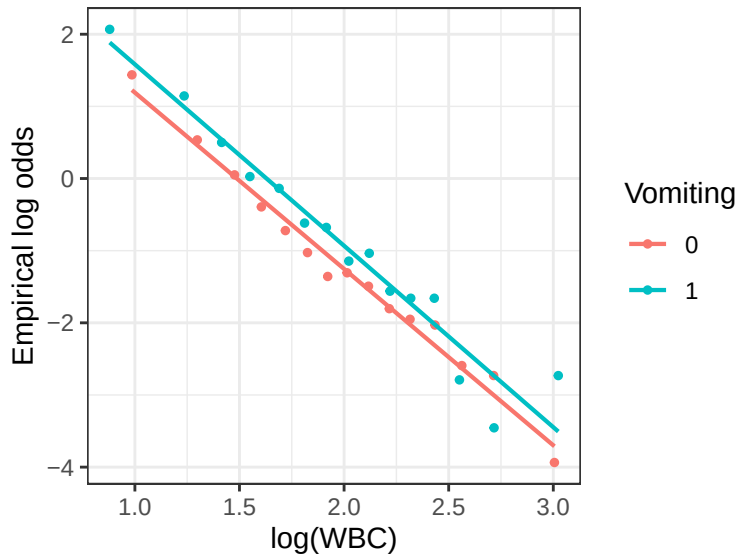


Does the shape assumption seem reasonable?

Binned empirical logit plots



Binned empirical logit plots



Limitations of empirical logit plots

- ▶ Does not assess a fitted model (performed *before* fitting the model)
- ▶ One point per *bin*, not one point per *observation*
- ▶ Can only plot one or two explanatory variables at a time

Randomized quantile residuals

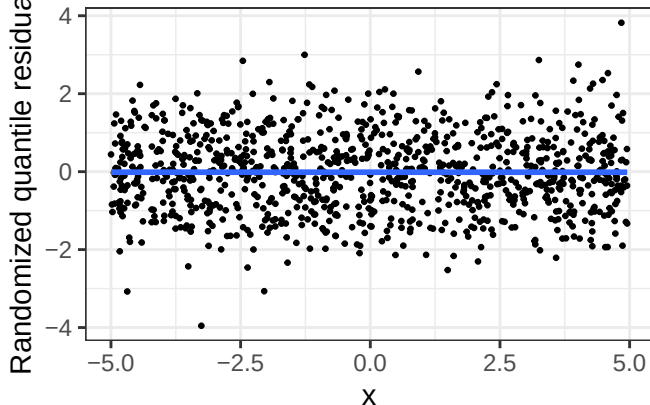
Define *randomized quantile residuals* $r_{Q,i}$:

Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i$

Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

Correctly specified model



Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i^2$

Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

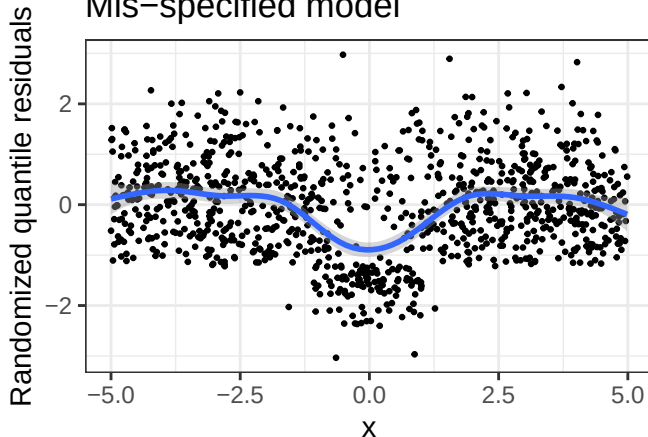
Question: What do you think the randomized quantile residual plot will look like?

Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i^2$

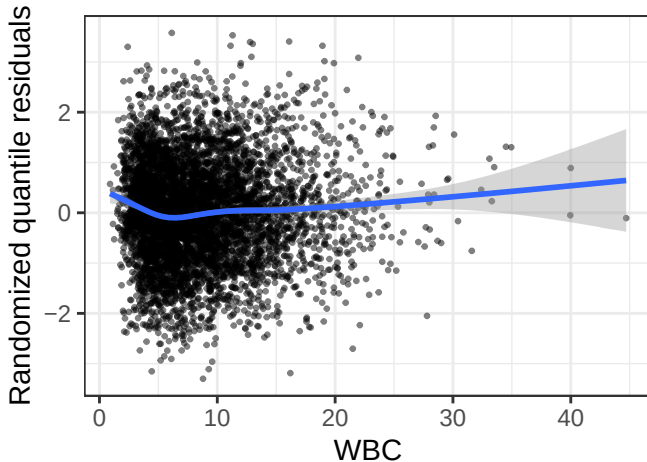
Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

Mis-specified model



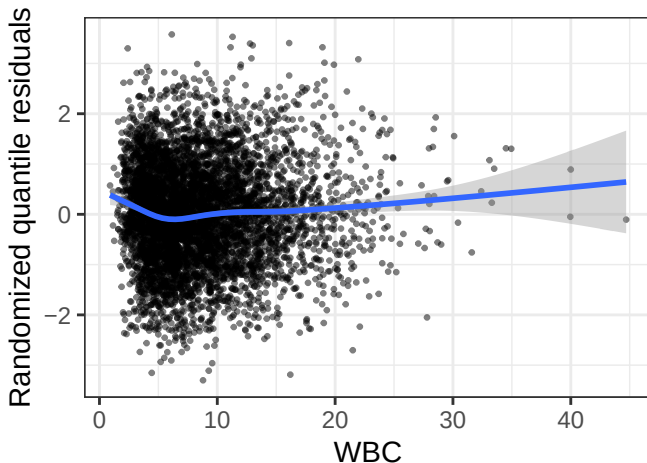
Example: dengue data

Model: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{WBC}_i$



Example: dengue data

Model: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{WBC}_i$



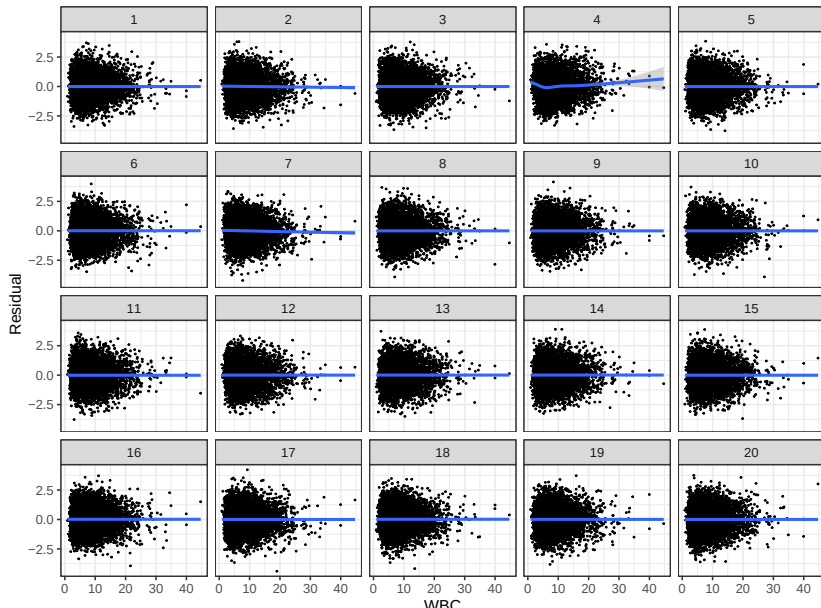
Question: How can we decide whether a plot is consistent with a correctly-specified model?

Parametric bootstrap

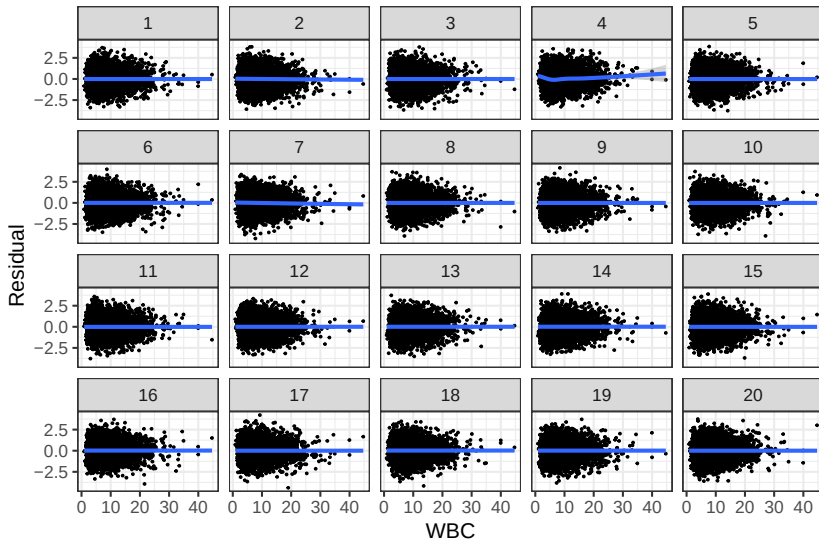
The *parametric bootstrap* generates data from the *fitted* model:

Visual inference

```
## decrypt("QUg2 qFyF Rx 8tLRyRtx Ko")
```



Visual inference



```
decrypt("QUg2 qFyF Rx 8tLRyRtx Ko")
```

```
## [1] "True data in position 4"
```